

The EverFlavor AI

Multi-Agent Diet & Discovery System

Data Report & Pipeline

Assignment: Phase 02: Data Pipeline and Exploratory Analysis (Weeks 4 -6)

Course: ITAI 2277 - Artificial Intel Resource

Institution: Houston City College

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Date: October 2, 2026

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GitHub: <https://github.com/Chloe-Tu2/The-EverFlavor-AI/tree/main>

1. Introduction

This report documents the data acquisition, preprocessing, feature engineering, and exploratory analysis for EverFlavor AI. EverFlavor AI is a multi-agent conversational system that generates authentic global recipes, tracks nutrition, respects dietary restrictions, and locates specialty ingredients. The work covers Weeks 4 through 6 of the course. It produces a documented and preprocessed dataset ready for later agent development.

The five target cuisine families for the prototype are Asian, European, Latin American, African, and Middle Eastern. All nutritional claims and ingredient data are grounded in public and verifiable sources rather than model-generated knowledge.

2. Data Acquisition and Collection (Week 4)

2.1 Identified Data Sources

1. Nutritional Data (Calories & Nutrients)

- USDA FoodData Central (official U.S. government)

- ❑ Official U.S. government food database and API. We will use it to retrieve and verify calorie and nutrient information for ingredients used in recipes.
- ❑ Website + Free API:
 - <https://fdc.nal.usda.gov/>
- ❑ **Open Food Facts (global, crowd-sourced)**
- ❑ Global, crowd-sourced food database and API. We will use it as an additional source for nutrition, ingredient, and packaged-food information.
- ❑ API Documentation:
 - <https://openfoodfacts.github.io/documentation/>

2. Recipe & International Cuisine Data

- ❑ **Food.com Recipes and User Interactions (Kaggle)**
- ❑ This will be our primary recipe dataset. It provides recipe and ingredient data that can be used for preprocessing, exploratory analysis, and model development.
- ❑ Kaggle Dataset:
 - <https://www.kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions>
- ❑ **CulinaryDB – Structured recipes across 22 world regions**
- ❑ This dataset will be used to explore ingredient relationships and cuisine patterns. It may also support the ingredient substitution and flavor-compatibility part of EverFlavor.
- ❑ Website:
 - <https://cosylab.iiitd.edu.in/culinarydb/>
- ❑ **Hugging Face – Recipes with Nutrition**
- ❑ This dataset will provide additional recipe information, including nutrition and diet/health-related information.
- ❑ Dataset:
 - <https://huggingface.co/datasets/datahiveai/recipes-with-nutrition>

3. Additional Dataset Considered

- ❑ **RecipeDB – 118,000+ recipes from 74 countries**

- This dataset contains international recipes from many countries. We will keep this as an additional option if more international recipe data is needed after evaluating the selected datasets.
- Website:
 - <https://cosylab.iitd.edu.in/recipedb>

4. Location & Sourcing Data

- **Google Places API**
- The Google Places API will be used by the Sourcing Agent to find real specialty grocery stores based on the user's location and cuisine needs.

2.2 Data Collection Methods

Collection was performed mainly in Google Colab notebooks and then committed to the project GitHub repository. The methods used are listed below.

- **USDA FoodData Central:** A free API key was obtained. Endpoint queries for foods and nutrients were made with the requests library. A sample bulk download of Foundation and SR Legacy datasets was also performed.
- **Open Food Facts:** The public API was used. No key is required for moderate volume. Product search and nutrition fields were retrieved through REST calls.
- **Food.com (Kaggle):** The dataset was downloaded with the official Kaggle API after authentication. Files were stored as raw CSV or Parquet.
- **Hugging Face recipes-with-nutrition:** The dataset was loaded with the datasets library using `load_dataset`.
- **CulinaryDB and RecipeDB:** These datasets were identified and documented as potential additional sources for later analysis of cuisine coverage, ingredient relationships, and flavor compatibility.

2.3 Data Storage and Organization

All project data follow a standard three-tier layout under the GitHub repository.

- `data/raw/` holds the original downloads and is never edited.

- data/interim/ holds cleaned intermediate files after initial quality filters.
- data/processed/ holds the final analysis-ready tables including train, validation, and test splits.

Supporting folders include notebooks/ for acquisition and EDA notebooks, src/ for reusable scripts, and docs/ for the data dictionary, sources list, and ethics notes. Large raw files are referenced from the repository README when size limits apply.

2.4 Privacy and Ethical Considerations

All datasets used are publicly available under open licenses or government open-data policies. The key points documented for the project are listed below.

1. No private or personally identifiable information is collected or stored by the data pipeline.
2. User dietary restrictions, remaining calories, and location remain session-based in the prototype and are not persisted without explicit consent.
3. Kaggle Food.com interaction data is treated as anonymized research data and terms of use are respected.
4. API rate limits and terms of service for USDA, Open Food Facts, and Google Places are followed.
5. No web scraping behind authentication walls is performed.

2.5 Initial Data Documentation

Documentation began in Week 4 and continued through Week 6. It includes the following files.

- docs/sources.md lists the source name, URL, license, access method, and access date for every dataset.
- docs/data_dictionary.md defines key fields such as recipe name, ingredients, cuisine or region, calories, macros, and allergens.
- docs/ethics_privacy.md records the privacy and ethical notes summarized above.

3. Data Preprocessing and Feature Engineering (Week 5)

3.1 Data Cleaning and Quality Assessment

Quality checks on the primary recipe and nutrition sources revealed several typical issues. These issues were addressed systematically.

- Missing or zero calorie and nutrient values appeared in some recipes or ingredients.
- Ingredient strings showed inconsistent formats including units, spelling variants, and quantity parsing problems.
- Duplicate or near-duplicate recipes were present.
- Cuisine or region labels were sometimes missing or too coarse.
- Occasional outlier values such as extreme calorie counts or unrealistic serving sizes were found.

Cleaning steps included dropping records with critical missing fields, standardizing ingredient text through lower-casing and unit normalization, removing exact duplicates, and flagging extreme outliers for review. Nutrition values from USDA and Open Food Facts were used to fill or verify missing calorie and macro fields when ingredient-level matching was possible.

3.2 Feature Engineering

Features were engineered specifically for EverFlavor AI needs.

1. Cuisine family label maps each recipe to one of the five target families (Asian, European, Latin American, African, Middle Eastern) or to Other.
2. Dietary restriction flags are binary indicators for common allergens and restrictions such as `contains_pork`, `contains_gluten`, `contains_dairy`, `vegetarian`, and `vegan`. These flags are derived from ingredient lists.
3. Nutrition features include calories per serving and available nutrient fields. Food.com provides calories directly, while several other nutrients are represented as percent daily values. USDA FoodData Central can be used for ingredient-level nutrient verification and future enrichment.

4. Ingredient list cleaned provides tokenized and normalized ingredient strings that support substitution logic and safety filtering.
5. Recipe length and complexity proxy uses the number of ingredients and estimated steps. This helps rank recipes when the user indicates low energy or limited time.

3.3 Data Transformation Pipelines

A simple and reproducible pipeline was implemented with notebooks and lightweight Python scripts. The steps are load raw tables, apply cleaning rules, join or enrich with nutrition lookups when feasible, generate engineered features, and write interim and processed Parquet or CSV files. The pipeline is documented so teammates can re-run or extend it.

3.4 Train / Validation / Test Splitting

The cleaned recipe set was split with a stratified approach by cuisine family when labels were available. The approximate ratio is 70 percent train, 15 percent validation, and 15 percent test. This supports future baseline retrieval or classification experiments while preserving cuisine representation across splits. The split indices and random seeds are recorded in the repository for reproducibility.

4. Exploratory Data Analysis (Week 6)

4.1 Statistical Analysis and Visualization

Summary statistics and visualizations were produced for the primary recipe and nutrition tables. Key views include the distribution of calories and available nutrition fields across recipes, coverage of the five target cuisine families, frequency of common ingredients and allergens, and the relationship between recipe complexity measured by ingredient count and calorie density.

4.2 Pattern Identification

Several patterns relevant to EverFlavor AI were identified.

1. Cuisine coverage is uneven across the current dataset. The Hugging Face data contains Asian, European, Latin American, and Middle Eastern recipes, but no recipes mapped to the African family. A portion of the recipes is also classified as Other. This supports keeping CulinaryDB and RecipeDB as potential additional sources to improve cuisine coverage.
2. Many recipes lack complete micronutrient data. Grounding calories and macro values against USDA and Open Food Facts is therefore essential for the Nutritionist Agent accuracy target of 5 percent error or less.
3. Ingredient name variation is high. A normalization step is required before reliable restriction filtering and substitution suggestions can be applied.
4. Shorter ingredient lists correlate with simpler preparation. This is useful for ranking recipes when the user indicates low energy or limited time.

4.3 Baseline Model Creation

A simple baseline was established for recipe retrieval under constraints. Recipes are first filtered by cuisine family and dietary restriction flags. Remaining recipes are then ranked by absolute difference from a target calorie value. This baseline provides a transparent reference against which later agent-based ranking and substitution logic can be compared. This rule-based baseline provides a reference for comparison with more advanced ranking or recommendation approaches in later phases.

4.4 Key Insights for the Multi-Agent System

- The Nutritionist Agent can use recipe-level nutrition data together with USDA and Open Food Facts lookups for additional verification and enrichment.
- The Chef Agent benefits from pre-computed restriction flags and cleaned ingredient lists to enforce 100 percent restriction compliance.
- The Sourcing Agent will remain independent of the recipe tables and will use Google Places within the Houston prototype boundary.
- Feature-engineered cuisine labels and complexity proxies enable better matching to user preferences such as mood, energy, and cuisine family.

5. Preprocessed Dataset Deliverable

The following artifacts constitute the preprocessed dataset package for this milestone.

- Cleaned recipe tables with engineered features including cuisine labels, dietary restriction flags, nutrition features, and cleaned ingredients. Cuisine-family labels are available in the processed Hugging Face dataset, while Food.com cuisine information is derived from recipe tags.
- Train, validation, and test splits that are stratified when possible.
- Supporting documentation including the data dictionary, sources list, and ethics note.
- Acquisition, cleaning, and EDA notebooks linked from the project GitHub repository.

All files are organized under the data hierarchy described in Section 2.3 and are referenced from the repository README.

6. Summary and Next Steps

Week 4 through 6 work established reliable public data sources, a clear storage layout, privacy safeguards, a cleaning and feature-engineering pipeline tailored to EverFlavor’s five cuisine families and restriction-safety requirements, and an initial exploratory analysis with a simple constraint-based baseline. The preprocessed dataset is ready to support Nutritionist and Chef Agent development, while the Sourcing Agent will be integrated separately using the Google Places API in a later phase.

Immediate next steps after Phase 02 are to connect USDA and Open Food Facts lookups into the Nutritionist Agent, apply the restriction flags inside the Chef Agent safety filter, and integrate Google Places for the Sourcing Agent within the Houston prototype boundary.

Appendix A - Source Summary Table

Source	Type	Primary Use	Access
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USDA FoodData Central	Nutrition API / DB	Calorie and nutrient verification	Free API + bulk
Open Food Facts	Nutrition API / DB	Global nutrition and ingredients	Public API
Food.com (Kaggle)	Recipe dataset	Primary recipes and ingredients	Kaggle download
CulinaryDB	Structured recipes	Cuisine patterns and substitution	Project website
HF Recipes with Nutrition	Recipe + nutrition	Additional recipes and diet tags	Hugging Face datasets
RecipeDB (optional)	International recipes	Extra country coverage	Project website
Google Places API	Location API	Specialty store sourcing	API key (Houston limit)